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Code:

```
1      [org 0x0100]
2      mov ax, 0xb800
3      mov es, ax
4      mov di, 0
5      nextchar: mov word [es:di], 0x0720
6      add di, 2
7      cmp di, 4000
8      jne nextchar
9      mov ax, 0x4c00
10     int 0x21
```

- 1) Address of memory address segment is loaded into the AX register

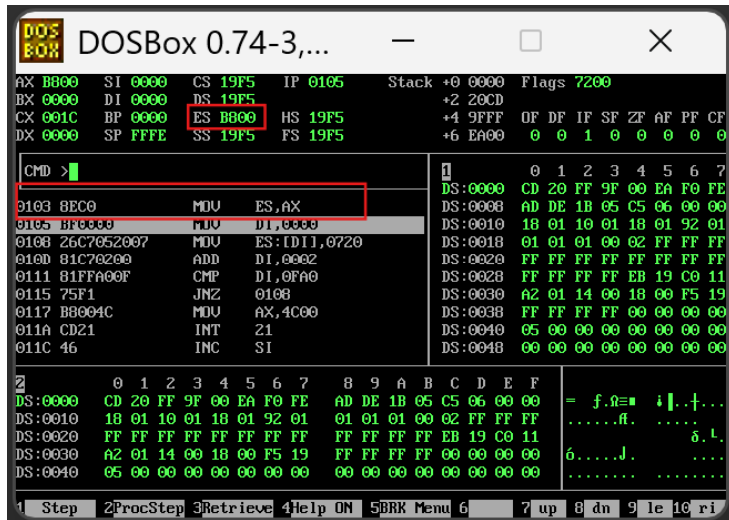
The screenshot shows the DOSBox 0.74-3 interface. The assembly code window displays the following instructions:

```
0100 B800B8 MOV AX,B800
0103 BEC0 MOV ES,AX
0105 BF0000 MOV DI,0000
0108 26C7052007 MOV ES:[DI],0720
010D 81C70200 ADD DI,0002
0111 81FA00F CMP DI,0FA0
0115 75F1 JNZ 0108
0117 B8004C MOV AX,4C00
011A CD21 INT 21
```

The memory dump window shows the following data:

Address	Hex	ASCII
DS:0000	CD 20 FF 9F 00 EA F0 FE	= f.~ = i . + ...
DS:0010	18 01 10 01 18 01 92 01	fl.
DS:0020	FF FF FF FF FF FF FF FF	δ. l.
DS:0030	A2 01 14 00 18 00 F5 19	6.....J.
DS:0040	05 00 00 00 00 00 00 00	

2) Value of AX (address of video address segment) is now stored in ES



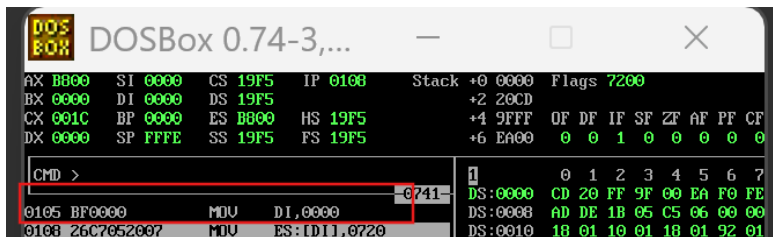
```
DOSBox 0.74-3,...
AX B800 SI 0000 CS 19F5 IP 0105 Stack +0 0000 Flags 7200
BX 0000 DI 0000 DS 19F5 +2 20CD
CX 001C BP 0000 ES B800 HS 19F5 +4 9FFF 0F DF IF SF ZF AF PF CF
DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 0 0 0 0

CMD >
0103 BEC0 MOV ES,AX
0105 BFB000 MOV DI,B000
0108 26C7652007 MOV ES:[DI],0720
010D 81C76200 ADD DI,0002
0111 81FFA00F CMP DI,0FA0
0115 75F1 JNZ 0108
0117 B8004C MOV AX,4C00
011A CD21 INT 21
011C 46 INC SI

DS:0000 CD 20 FF 9F 00 EA F0 FE AD DE 1B 05 C5 06 00 00
DS:0010 18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF
DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF EB 19 C0 11
DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00
DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

3) 0000 is moved in DI to make sure it stores no existing value and will be used as a offset

0000 now points to the first character on the screen



```
DOSBox 0.74-3,...
AX B800 SI 0000 CS 19F5 IP 0108 Stack +0 0000 Flags 7200
BX 0000 DI 0000 DS 19F5 +2 20CD
CX 001C BP 0000 ES B800 HS 19F5 +4 9FFF 0F DF IF SF ZF AF PF CF
DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 0 0 0 0

CMD >
0105 BFB000 MOV DI,0000
0108 26C7652007 MOV ES:[DI],0720

DS:0000 CD 20 FF 9F 00 EA F0 FE AD DE 1B 05 C5 06 00 00
DS:0010 18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF
DS:0020 18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF
```

- 4) 0x720 represents attribute and character, where 0x07 represent attribute (black text on black background) and 0x20 represent character. So this writes a black space at the screen location.

DOSBox 0.74-3,...

AX	B800	SI	0000	CS	19F5	IP	010D	Stack	+0 0000	Flags	7200
BX	0000	DI	0000	DS	19F5				+2 20CD		
CX	001C	BP	0000	ES	B800	HS	19F5		+4 9FFF	OF	DF IF SF ZF AF PF CF
DX	0000	SP	FFFE	SS	19F5	FS	19F5		+6 EA00	0	0 1 0 0 0 0 0

CMD > █

0108	26C7652007	MOV	ES:DI, 0720	DS:0000	CD 20 FF 9F 00 EA F0 FE		
010D	81C76200	ADD	DI, 0002	DS:0008	AD DE 1B 05 C5 06 00 00		
0111	81FFA00F	CMF	DI, 0FA0	DS:0010	18 01 10 01 18 01 92 01		
0115	75F1	JNZ	0108	DS:0018	01 01 01 00 02 FF FF FF		
0117	B8004C	MOV	AX, 4C00	DS:0020	FF FF FF FF FF FF FF FF		
011A	CD21	INT	21	DS:0028	FF FF FF FF EB 19 C0 11		
011C	46	INC	SI	DS:0030	A2 01 14 00 18 00 F5 19		
011D	E6C7	OUT	[C7], AL	DS:0038	FF FF FF FF 00 00 00 00		
011F	46	INC	SI	DS:0040	05 00 00 00 00 00 00 00		
0120	F60000	TEST	IBX, SI, 00	DS:0048	00 00 00 00 00 00 00 00		

1 Step 2 ProcStep 3 Retrieve 4 Help ON 5 BRK Menu 6 7 up 8 dn 9 le 10 ri

- 5) Adds 2 bytes in the offset as for each block (pixel) 1 for character and 1 for the attribute . so DI increments by 2 in each iteration

DOSBox 0.74-3,...

AX	B800	SI	0000	CS	19F5	IP	0111	Stack	+0 0000	Flags	7200
BX	0000	DI	0002	DS	19F5				+2 20CD		
CX	001C	BP	0000	ES	B800	HS	19F5		+4 9FFF	OF	DF IF SF ZF AF PF CF
DX	0000	SP	FFFE	SS	19F5	FS	19F5		+6 EA00	0	0 1 0 0 0 0 0

CMD > █

0108	26C7652007	MOV	ES:DI, 0720	DS:0000	CD 20 FF 9F 00 EA F0 FE		
010D	81C76200	ADD	DI, 0002	DS:0008	AD DE 1B 05 C5 06 00 00		
0111	81FFA00F	CMF	DI, 0FA0	DS:0010	18 01 10 01 18 01 92 01		
0115	75F1	JNZ	0108	DS:0018	01 01 01 00 02 FF FF FF		
0117	B8004C	MOV	AX, 4C00	DS:0020	FF FF FF FF FF FF FF FF		
011A	CD21	INT	21	DS:0028	FF FF FF FF EB 19 C0 11		
011C	46	INC	SI	DS:0030	A2 01 14 00 18 00 F5 19		
011D	E6C7	OUT	[C7], AL	DS:0038	FF FF FF FF 00 00 00 00		
011F	46	INC	SI	DS:0040	05 00 00 00 00 00 00 00		
0120	F60000	TEST	IBX, SI, 00	DS:0048	00 00 00 00 00 00 00 00		

1 Step 2 ProcStep 3 Retrieve 4 Help ON 5 BRK Menu 6 7 up 8 dn 9 le 10 ri

- 6) Screen has $80 \times 25 = 2000$ cells. Each cell = 2 bytes thus $2000 \times 2 = 4000$ bytes.
So when di is equal to 4000 (0FA0h), the entire screen is cleared.

```

DOSBox 0.74-3,...
AX:0000 SI:0000 CS:19F5 IP:0115 Stack:+0 0000 Flags:7201
BX:0000 DI:0002 DS:19F5      +2 20CD
CX:001C BP:0000 ES:B000 HS:19F5 +4 9FFF OF DF IF SF ZF AF FF CF
DX:0000 SP:FFFF SS:19F5 FS:19F5 +6 E400 0 0 1 1 0 0 0 1

CMD >
0114:011FA00F CMP     DI,0FA0h
0115:7571     JNZ     0103
0117:B0004C    MOV     AX,4C00
011A:CD21     INT     21
011C:46       INC     SI
011D:E6C7     OUT     (CX),AL
011F:46       INC     SI
0120:F60000    TEST    [BX+SI],00
0123:8B46F6    MOV     AX,[BP-0A]

DS:0000  0 1 2 3 4 5 6 7 8 9 A B C D E F  = f.8.  i.+.
DS:0010  CD 20 FF 9F 00 EA F0 FE  AD DE 1B 05 C5 06 00 00
DS:0020  1B 01 10 01 1B 01 92 01  01 01 01 00 02 FF FF FF
DS:0030  FF FF FF FF FF FF FF FF  FF FF FF FF EB 19 C0 11
DS:0040  02 01 14 00 1B 00 F5 19  FF FF FF FF 00 00 00 00
  
```

- 7) This loop make sure the program completes all it iteration in this case 4000 if not it will repeat the process

```

DOSBox 0.74-3,...
AX:0000 SI:0000 CS:19F5 IP:0115 Stack:+0 0000 Flags:7201
BX:0000 DI:0002 DS:19F5      +2 20CD
CX:001C BP:0000 ES:B000 HS:19F5 +4 9FFF OF DF IF SF ZF AF FF CF
DX:0000 SP:FFFF SS:19F5 FS:19F5 +6 E400 0 0 1 1 0 0 0 1

CMD >
0114:011FA00F CMP     DI,0FA0h
0115:7571     JNZ     0103
0117:B0004C    MOV     AX,4C00
011A:CD21     INT     21
011C:46       INC     SI
011D:E6C7     OUT     (CX),AL
011F:46       INC     SI
0120:F60000    TEST    [BX+SI],00
0123:8B46F6    MOV     AX,[BP-0A]

DS:0000  0 1 2 3 4 5 6 7 8 9 A B C D E F  = f.8.  i.+.
DS:0010  CD 20 FF 9F 00 EA F0 FE  AD DE 1B 05 C5 06 00 00
DS:0020  1B 01 10 01 1B 01 92 01  01 01 01 00 02 FF FF FF
DS:0030  FF FF FF FF FF FF FF FF  FF FF FF FF EB 19 C0 11
DS:0040  02 01 14 00 1B 00 F5 19  FF FF FF FF 00 00 00 00
  
```

--Program is terminated--